

Step1: Import Necessary Libraries

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import nltk
import re

nltk.download("stopwords")
nltk.download("punkt")
nltk.download("wordnet")
nltk.download("omw-1.4")
nltk.download("punkt_tab")

[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.

True
```

Step2: Load Datasets

```
from google.colab import files
uploaded = files.upload()

<IPython.core.display.HTML object>

Saving IMDB Dataset.csv to IMDB Dataset.csv

import pandas as pd
import io
df = pd.read_csv(io.BytesIO(uploaded['IMDB Dataset.csv']))
df.head()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 50000,\n  \"fields\": [\n    {\n      \"column\": \"review\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 49582,\n        \"samples\": [\n          \"\"\"Soul Plane\"\"\" is a horrible attempt at comedy that only should appeal people with thick skulls, bloodshot eyes and furry pawns. <br /><br />The plot is not only incoherent but also non-existent, acting is mostly sub sub-par with a gang of highly moronic and dreadful characters thrown in for bad measure, jokes are
```

often spotted miles ahead and almost never even a bit amusing. This movie lacks any structure and is full of racial stereotypes that must have seemed old even in the fifties, the only thing it really has going for it is some pretty ladies, but really, if you want that you can rent something from the \\\\"Adult\\\\" section. OK?

I can hardly see anything here to recommend since you'll probably have a lot a better and productive time chasing rats with a sledgehammer or inventing waterproof teabags or whatever.

2/10\\",\n

\"Guest from the Future tells a fascinating story of time travel, friendship, battle of good and evil -- all with a small budget, child actors, and few special effects. Something for Spielberg and Lucas to learn from. ;) A sixth-grader Kolya \\\\"Nick\\\\" Gerasimov finds a time machine in the basement of a decrepit building and travels 100 years into the future. He discovers a near-perfect, utopian society where robots play guitars and write poetry, everyone is kind to each other and people enjoy everything technology has to offer. Alice is the daughter of a prominent scientist who invented a device called Mielophone that allows to read minds of humans and animals. The device can be put to both good and bad use, depending on whose hands it falls into. When two evil space pirates from Saturn who want to rule the universe attempt to steal Mielophone, it falls into the hands of 20th century school boy Nick. With the pirates hot on his tracks, he travels back to his time, followed by the pirates, and Alice. Chaos, confusion and funny situations follow as the luckless pirates try to blend in with the earthlings. Alice enrolls in the same school Nick goes to and demonstrates superhuman abilities in PE class. The catch is, Alice doesn't know what Nick looks like, while the pirates do. Also, the pirates are able to change their appearance and turn literally into anyone. (Hmm, I wonder if this is where James Cameron got the idea for Terminator...) Who gets to Nick -- and Mielophone -- first? Excellent plot, non-stop adventures, and great soundtrack. I wish Hollywood made kid movies like this one...\\",\n

\"\\\\"National Treasure\\\\" (2004) is a thoroughly misguided hodge-podge of plot entanglements that borrow from nearly every cloak and dagger government conspiracy clich\\u00e9 that has ever been written. The film stars Nicholas Cage as Benjamin Franklin Gates (how precious is that, I ask you?); a seemingly normal fellow who, for no other reason than being of a lineage of like-minded misguided fortune hunters, decides to steal a 'national treasure' that has been hidden by the United States founding fathers. After a bit of subtext and background that plays laughably (unintentionally) like Indiana Jones meets The Patriot, the film degenerates into one misguided whimsy after another \\u0096 attempting to create a 'Stanley Goodspeed' regurgitation of Nicholas Cage and launch the whole convoluted mess forward with a series of high octane, but disconnected misadventures.

The relevancy and logic to having George Washington and his motley crew of patriots burying a king's ransom someplace on native soil, and then, going through the meticulous plan of leaving clues scattered throughout U.S. currency art work, is

something that director Jon Turteltaub never quite gets around to explaining. Couldn't Washington found better usage for such wealth during the start up of the country? Hence, we are left with a mystery built on top of an enigma that is already on shaky ground by the time Ben appoints himself the new custodian of this untold wealth. Ben's intentions are noble \u0096 if confusing. He's set on protecting the treasure. For who and when?\u0085your guess is as good as mine.

But there are a few problems with Ben's crusade. First up, his friend, Ian Holmes (Sean Bean) decides that he can't wait for Ben to make up his mind about stealing the Declaration of Independence from the National Archives (oh, yeah \u0096 brilliant idea!). Presumably, the back of that famous document holds the secret answer to the ultimate fortune. So Ian tries to kill Ben. The assassination attempt is, of course, unsuccessful, if overly melodramatic. It also affords Ben the opportunity to pick up, and pick on, the very sultry curator of the archives, Abigail Chase (Diane Kruger). She thinks Ben is clearly a nut \u0096 at least at the beginning. But true to action/romance form, Abby's resolve melts quicker than you can say, \\\

"is that the Hope Diamond?\\\" The film moves into full X-File-ish mode, as the FBI, mistakenly believing that Ben is behind the theft, retaliate in various benign ways that lead to a multi-layering of action sequences reminiscent of Mission Impossible meets The Fugitive. Honestly, don't those guys ever get 'intelligence' information that is correct? In the final analysis, \\\

"National Treasure\\\" isn't great film making, so much as it's a patchwork rehash of tired old bits from other movies, woven together from scraps, the likes of which would make IL' Betsy Ross blush.

The Buena Vista DVD delivers a far more generous treatment than this film is deserving of. The anamorphic widescreen picture exhibits a very smooth and finely detailed image with very rich colors, natural flesh tones, solid blacks and clean whites. The stylized image is also free of blemishes and digital enhancements. The audio is 5.1 and delivers a nice sonic boom to your side and rear speakers with intensity and realism. Extras include a host of promotional junket material that is rather deep and over the top in its explanation of how and why this film was made. If only, as an audience, we had had more clarification as to why Ben and co. were chasing after an illusive treasure, this might have been one good flick. Extras conclude with the theatrical trailer, audio commentary and deleted scenes. Not for the faint-hearted \u0096 just the thick-headed.\\\"\\n

```

    ],\n    \"semantic_type\": \"\", \n    \"description\": \"\"\\n    },\n    {\n    \"column\": \n    \"sentiment\", \n    \"properties\": {\n    \"dtype\": \n    \"category\", \n    \"num_unique_values\": 2, \n    \"samples\": \n    [\n    \"negative\", \n    \"positive\"\\n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\"\\n    } \n    } \n    ],\n    \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```
df.shape
```

```
(50000, 2)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 50000 entries, 0 to 49999  
Data columns (total 2 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   review      50000 non-null   object  
1   sentiment    50000 non-null   object  
dtypes: object(2)  
memory usage: 781.4+ KB
```

```
df.describe()
```

```
{"summary":{"name": "df", "rows": 4, "fields": [ {  
  "column": "review", "properties": {  
    "dtype": "string", "num_unique_values": 4, "samples": [ 49582, "5", "50000"  
    ], "semantic_type": "",  
    "description": "" } }, {  
    "column": "sentiment", "properties": {  
      "dtype": "string", "num_unique_values": 4, "samples": [ 2, "25000", "50000"  
      ], "semantic_type": "",  
      "description": "" } } ] }, "type": "dataframe"}
```

```
df.columns
```

```
Index(['review', 'sentiment'], dtype='object')
```

Missing Value Check

```
df.isnull().sum()
```

```
review      0  
sentiment    0  
dtype: int64
```

Duplicate Value Check

```
df.duplicated().sum()
```

```
np.int64(418)
```

Remove Duplicated Values

```
df.drop_duplicates(inplace=True)
```

Step3: Text Cleaning

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
```

Initialize stopword and lemmatizer

```
stop_words=set(stopwords.words("english"))
lemmatizer=WordNetLemmatizer()
```

Creat the preprocessing Function

```
def preprocess_text(text):
    # 1.Text Cleaning and Normalization
    text=text.lower()                # Convert to lowercase
    text=re.sub(r"<.*?", "", text)    # Remove HTML tags
    text=re.sub(r"http\S+|www\S+", "", text) # Remove URLs
    text=re.sub(r"[^a-zA-Z\s]", "", text) # Remove punctuation &
    numbers                           # Remove extra whitespaces
    text=re.sub(r"\s+", "", text).strip()

    #2.Tokenization
    tokens=word_tokenize(text)

    #3.Stopword Removal
    tokens=[word for word in tokens if word not in stop_words]

    # 4. Lemmatization
    tokens = [lemmatizer.lemmatize(word) for word in tokens]

    #5.Join tokens back into a sentence
    return "".join(tokens)
```

```
import nltk
```

```
nltk.download('punkt', force=True)
nltk.download('punkt_tab', force=True)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

```
True
```

```
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
```

```
df["clean_review"] = df["review"].apply(preprocess_text)
```

```
df[["review", "clean_review"]].head()
```

```

{"summary": "{\n  \"name\":\n  \"df[[\\\"review\\\"],\\\"clean_review\\\"]\\\", \n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"review\", \n      \"properties\": {\n        \"dtype\": \"string\", \n        \"num_unique_values\": 5,\n        \"samples\": [\n          \"A wonderful little production. <br /><br />The filming technique is very unassuming- very old-time-BBC fashion and gives a comforting, and sometimes discomforting, sense of realism to the entire piece. <br /><br />The actors are extremely well chosen- Michael Sheen not only \\\"has got all the polari\\\" but he has all the voices down pat too! You can truly see the seamless editing guided by the references to Williams' diary entries, not only is it well worth the watching but it is a terrifically written and performed piece. A masterful production about one of the great master's of comedy and his life. <br /><br />The realism really comes home with the little things: the fantasy of the guard which, rather than use the traditional 'dream' techniques remains solid then disappears. It plays on our knowledge and our senses, particularly with the scenes concerning Orton and Halliwell and the sets (particularly of their flat with Halliwell's murals decorating every surface) are terribly well done.\", \n          \"Petter Mattei's \\\"Love in the Time of Money\\\" is a visually stunning film to watch. Mr. Mattei offers us a vivid portrait about human relations. This is a movie that seems to be telling us what money, power and success do to people in the different situations we encounter. <br /><br />This being a variation on the Arthur Schnitzler's play about the same theme, the director transfers the action to the present time New York where all these different characters meet and connect. Each one is connected in one way, or another to the next person, but no one seems to know the previous point of contact. Stylishly, the film has a sophisticated luxurious look. We are taken to see how these people live and the world they live in their own habitat.<br /><br />The only thing one gets out of all these souls in the picture is the different stages of loneliness each one inhabits. A big city is not exactly the best place in which human relations find sincere fulfillment, as one discerns is the case with most of the people we encounter.<br /><br />The acting is good under Mr. Mattei's direction. Steve Buscemi, Rosario Dawson, Carol Kane, Michael Imperioli, Adrian Grenier, and the rest of the talented cast, make these characters come alive.<br /><br />We wish Mr. Mattei good luck and await anxiously for his next work.\", \n          \"I thought this was a wonderful way to spend time on a too hot summer weekend, sitting in the air conditioned theater and watching a light-hearted comedy. The plot is simplistic, but the dialogue is witty and the characters are likable (even the well bread suspected serial killer). While some may be disappointed when they realize this is not Match Point 2: Risk Addiction, I thought it was proof that Woody Allen is still fully in control of the style many of us have grown to love.<br /><br />This was the most I'd laughed at one of Woody's comedies in years (dare I say a decade?). While I've never been impressed with Scarlet Johanson, in this she managed to tone down her
        ]
      }
    ]
  }

```

```

\\\\"sexy\\\\" image and jumped right into a average, but spirited young
woman.<br /><br />This may not be the crown jewel of his career, but
it was wittier than \\\"Devil Wears Prada\\\" and more interesting
than \\\"Superman\\\" a great comedy to go see with friends.\\\"\\n
],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n    },\\n    {\\n      \\\"column\\\": \\\"clean_review\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 5,\\n      \\\"samples\\\": [\\n
\\\"awonderfullittleproductionbrbrthefilmingtechniqueisveryunassumingver
yoldtimebbcfashionandgivesacomfortingandsometimesdiscomfortingsenseofr
ealismtotheentirepiecebrbrtheactorsareextremelywellchosenmichaelsheenn
otonlyhasgotallthepolaributhehasallthevoicesdownpattoooucantrulyseeth
eseamlesseditingguidedbythereferencestowilliamsdiaryentriesnotonlyisit
wellworththewatchingbutitisaterrificlywrittenandperformedpieceamasterf
ulproductionaboutoneofthegreatmastersofcomedyandhislifebrbrtherealismr
eallycomeshomewiththelittlethingsthefantasyoftheguardwhichratherthanu
sethetraditionaldreamtechniquesremainssolidthendisappearsitplaysonourkn
owledgeandoursensesparticularlywiththescenesconcerningortonandhalliwe
llandthesetsparticularlyoftheirflatwithhalliwellsmuralsdecoratingevery
surfaceareterriblywelldone\\\",\\n
\\\"pettermatteisloveinthetimeofmoneyisavisuallystunningfilmtowatchmrm
atteioffersusavividportraitabouthumanrelationsthisismoviehatseemstobet
ellinguswhatmoneypowerandsuccessdotopeopleinthedifferentsituationswee
ncounterbrbrthisbeingavariationontheearthurschnitzlersplayaboutthesame
themethedirectortransferstheactiontothepresenttimenewyorkwhereallthesedi
fferentcharactersmeetandconnecteachoneisconnectedinonewayoranothertoth
enextpersonbutnooneseemstoknowthepreviouspointofcontactstylishlythefil
mhasasophisticatedluxuriouslookwearetakentoseehowthesepeopleliveandthe
worldtheyliveintheirownhabitatbrbrtheonlythingonegetsoutofallthesesoul
sinthepictureisthedifferentstagesoflonelinesseachoneinhabitsabigcityis
notexactlythebestplaceinwhichhumanrelationsfindsincerefulfillmentasone
discernsisthecasewithmostofthepeopleweencounterbrbrtheactingisgoodunde
rmmatteisdirectionstevebuscemirosariodawsoncarolkanemichaelimperiolia
driangrenierandtherestofthetalentedcastmakethesecharacterscomealivebr
rwewishmrmatteigoodluckandawaitanxiouslyforhisnextwork\\\",\\n
\\\"ithoughtthiswasawonderfulwaytospendtimeonatoohotsummerweekendsitting
intheairconditionedtheaterandwatchingalightheartedcomedytheplotissimpl
isticbutthedialogueiswittyandthecharactersarelikableeventhewellbreadsu
spectedserialkillerwhilesomesmaybedisappointedwhentheyrealizethisisnotm
atchpointriskaddictionithoughtitwasproofthatwoodyallenisstillfullyinco
ntrolofthestylemanyofushavegrowntolovebrbrthiswasthemostidlaughedatone
ofwoodyscomediesinyearsdareisayadecadewhileivehavebeenimpressedwithsc
arletjohansoninthisshemanagedtotonedownhersexyimageandjumpedrightintoa
averagebutspiritedyoungwomanbrbrthismaynotbethecrownjewelofhiscareerbu
titwaswittierthandevilwearspradaandmoreinterestingthansupermanagreatco
medytogoseewithfriends\\\"\\n      ],\\n      \\\"semantic_type\\\":
\\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n    }\\n  ]\\n
n}\\\", \"type\": \"dataframe\"}

```

Step4:Baseline Sentiment Classifier Using TF-IDF

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score,
confusion_matrix,classification_report
```

Define Features and target

```
df["label"]=df["sentiment"].map({
    "positive":1,
    "negative":0
})

X=df["clean_review"]
y=df["label"]
```

Train-Test Split

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42,stratify=y)
```

convert text into TF-IDF vectorizer

```
tfidf=TfidfVectorizer(
    max_features=5000,
    stop_words="english",
    ngram_range=(1,2)
)

X_train_tfidf=tfidf.fit_transform(X_train)
X_test_tfidf=tfidf.transform(X_test)
```

Train the Baseline Classifier

```
model=LogisticRegression(max_iter=1000)
model.fit(X_train_tfidf,y_train)

LogisticRegression(max_iter=1000)
```

Predict Sentiment

```
y_pred=model.predict(X_test_tfidf)
```

Evaluate The Model

Accuracy

```
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
```

Accuracy: 0.5018654835131592

Confusion Matrix

```
cm=confusion_matrix(y_test,y_pred)
print(cm)
```

```
[[ 0 4940]
 [ 0 4977]]
```

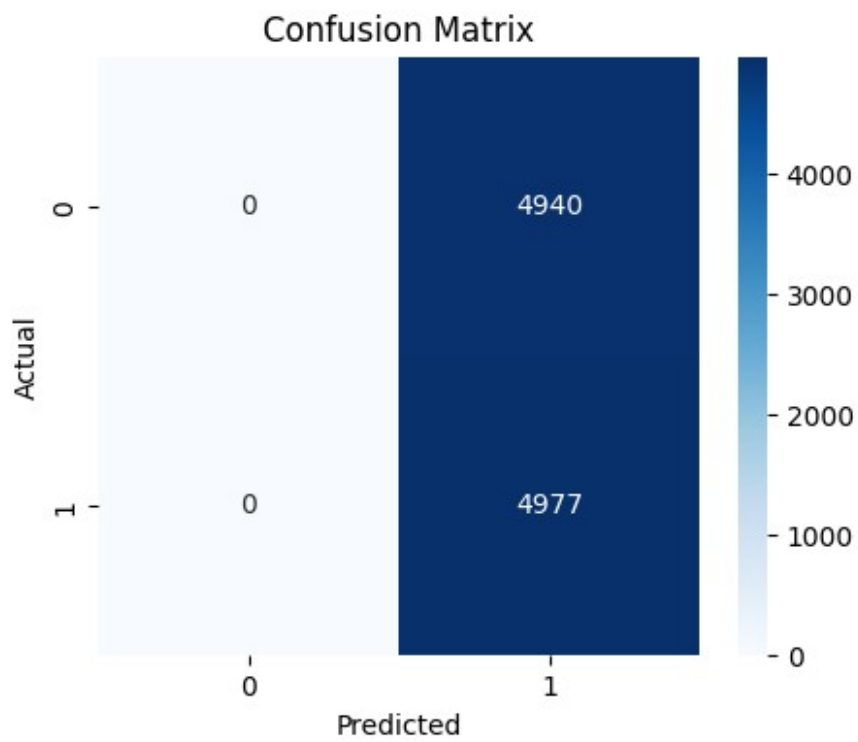
To Visualize it

```
plt.figure(figsize=(5,4))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues"
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```



Classification Report

```
print(classification_report(y_test,y_pred))
```

	precision	recall	f1-score	support
0	0.00	0.00	0.00	4940
1	0.50	1.00	0.67	4977
accuracy			0.50	9917
macro avg	0.25	0.50	0.33	9917
weighted avg	0.25	0.50	0.34	9917

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
```

Step5:Derive emotion vectors using word embedding and cosine similarity

install gensim

```
pip install gensim
```

Collecting gensim

```
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in
/usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in
/usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in
```

```
/usr/local/lib/python3.12/dist-packages (from gensim) (8.0.0)
Requirement already satisfied: wrapt in
/usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1-
>gensim) (2.2.2)
Downloading gensim-4.4.0-cp312-cp312-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
===== 27.9/27.9 MB 58.7 MB/s eta
0:00:00

Successfully installed gensim-4.4.0

import gensim.downloader as api
from sklearn.metrics.pairwise import cosine_similarity

embedding_model=api.load("glove-wiki-gigaword-100")

[=====] 100.0%
128.1/128.1MB downloaded
```

Define Emotion Keywords

```
emotion_words = {

    "joy": [
        "happy", "joy", "love", "excellent", "amazing",
        "wonderful", "fantastic", "delight"
    ],

    "anger": [
        "angry", "hate", "furious", "annoying",
        "terrible", "awful", "rage"
    ],

    "sadness": [
        "sad", "cry", "depressed",
        "lonely", "heartbreaking", "pain"
    ],

    "fear": [
        "fear", "scary", "terror",
        "horror", "frightening", "panic"
    ],

    "surprise": [
        "surprise", "unexpected",
        "astonishing", "shocking"
    ],

    "disgust": [
        "disgust", "gross",
        "boring", "horrible",
```

```
        "disappointing"  
    ]  
}
```

Create a Function to Generate a Review Embedding

```
def get_review_vector(text):  
    vectors = []  
    for word in text.split():  
        if word in embedding_model:  
            vectors.append(embedding_model[word])  
    if len(vectors) == 0:  
        return np.zeros(embedding_model.vector_size)  
    return np.mean(vectors, axis=0)
```

Create Emotion Vectors

```
emotion_vectors = {}  
for emotion, words in emotion_words.items():  
    vectors = []  
    for word in words:  
        if word in embedding_model:  
            vectors.append(embedding_model[word])  
    emotion_vectors[emotion] = np.mean(vectors, axis=0)
```

Calculate Cosine Similarity

```
def emotion_scores(review):  
    review_vector = get_review_vector(review)  
    scores = {}  
    for emotion, emotion_vector in emotion_vectors.items():  
        similarity = cosine_similarity(  
            review_vector.reshape(1, -1),  
            emotion_vector.reshape(1, -1)  
        )[0][0]
```

```
scores[emotion] = round(similarity,4)

return scores
```

Apply to the Dataset

```
df["emotion_scores"] = df["clean_review"].apply(emotion_scores)
```

Convert Scores into Columns

```
emotion_df = df["emotion_scores"].apply(pd.Series)

df = pd.concat([df, emotion_df], axis=1)
```

Predict the Dominant Emotion

```
emotion_columns = [
    "joy",
    "anger",
    "sadness",
    "fear",
    "surprise",
    "disgust"
]

df["predicted_emotion"] = df[emotion_columns].idxmax(axis=1)
```

View Results

```
df[[
    "clean_review",
    "joy",
    "anger",
    "sadness",
    "fear",
    "surprise",
    "disgust",
    "predicted_emotion"
]].head()

{"summary":{"\n  \"name\": \"\"]\n\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"clean_review\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \"awonderfullittleproductionbrbrthefilmingtechniqueisveryunassumingver\n          yoldtimebbcfashionandgivesacomfortingandsometimesdiscomfortingsenseofr\n          ealismtotheentirepiecebrbrtheactorsareextremelywellchosenmichaelsheenn
```

otonlyhasgotallthepolaributhehasallthevoicesdownpattooyoucantrulyseeth
eseamlesseditingguidedbythereferencestowilliamsdiaryentriesnotonlyisit
wellworththewatchingbutitisaterrificlywrittenandperformedpieceamasterf
ulproductionaboutoneofthegreatmastersofcomedyandhislifebrbrtherealismr
eallycomeshomewiththelittlethingsthefantasyoftheguardwhichratherthan
ethetraditionaldreamtechniquesremainssolidthendisappearsitplaysonourkn
owledgeandoursensesparticularlywiththescenesconcerningortonandhalliwe
landthesetsparticularlyoftheirflatwithhalliwellsmuralsdecoratingevery
urfaceareterriblywelldone",\n

\n"pettermatteisloveinthetimeofmoneyisavisuallystunningfilmtowatchmrm
teioffersusavividportraitabouthumanrelationsthisismoviethatseemstobet
ellinguswhatmoneypowerandsuccessdotopeopleinthedifferentsituationsween
counterbrbrthisbeingavariationontheearthurschnitzlersplayaboutthesameth
emethedirectortransferstheactiontothepresenttimenewyorkwhereallthesedi
fferentcharactersmeetandconnecteachoneisconnectedinonewayoranothertoth
enextpersonbutnooneseemstoknowthepreviouspointofcontactstylishlythefil
mhasasophisticatedluxuriouslookwearetakentoseehowthesepeopleliveandthe
worldtheyliveintheirownhabitatbrbrtheonlythingonegetsoutofallthesesoul
sinthepictureisthedifferentstagesoflonelinesseachoneinhabitsabigcityis
notexactlythebestplaceinwhichhumanrelationsfindsincerefulfillmentasone
discernsisthecasewithmostofthepeopleweencounterbrbrtheactingisgoodunde
rmrmatteisdirectionstevebuscemirosariodawsoncarolkanemichaelimperiolia
driangrenierandtherestofthetalentedcastmakethesecharacterscomealivebr
rwewishmrmattegoodluckandawaitanxiouslyforhisnextwork",\n

\n"ithoughtthiswasawonderfulwaytospendtimeonatoohtsummerweekendsitting
intheairconditionedtheaterandwatchingalightheartedcomedytheplotissimpl
isticbutthedialogueiswittyandthecharactersarelikableeventhewellbreadsu
spectedserialkillerwhilesomemaybedisappointedwhentheyrealizethisisnotm
atchpointriskaddictionithoughtitwasproofthatwoodyallenisstillfullyinco
ntrolofthestylemanyofushavegrowntolovebrbrthiswasthemostidlaughedatone
ofwoodyscomediesinyearsdareisayadecadewhilei'veneverbeenimpressedwiths
carletjohansoninthisshemanagedtotonedownhersexyimageandjumpedrightinto
averagebutspiritedyoungwomanbrbrthismaynotbethecrownjewelofhiscareerbu
titwaswittierthandevilwearspradaandmoreinterestingthansupermanagreatco
medytogoseewithfriends",\n

```
],\n    \"semantic_type\":\n    \"\", \n    \"description\": \"\", \n    },\n    {\n    \"column\": \"joy\", \n    \"properties\": {\n    \"dtype\":\n    \"number\", \n    \"std\": 0.0, \n    \"min\": 0.0, \n    \"max\": 0.0, \n    \"num_unique_values\": 1, \n    \"samples\":\n    [\n    0.0\n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\", \n    },\n    {\n    \"column\":\n    \"anger\", \n    \"properties\": {\n    \"dtype\": \"number\", \n    \"std\": 0.0, \n    \"min\": 0.0, \n    \"max\": 0.0, \n    \"num_unique_values\": 1, \n    \"samples\": [\n    0.0\n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\", \n    },\n    {\n    \"column\": \"sadness\", \n    \"properties\": {\n    \"dtype\": \"number\", \n    \"std\":\n    0.0, \n    \"min\": 0.0, \n    \"max\": 0.0, \n    \"num_unique_values\": 1, \n    \"samples\": [\n    0.0\n
```

```

],\n      \"semantic_type\": \"\", \n      \"description\": \"\"\n}\n    },\n    {\n      \"column\": \"fear\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 0.0, \n        \"min\": 0.0, \n        \"max\": 0.0, \n        \"num_unique_values\": 1, \n        \"samples\": [\n          0.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"surprise\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 0.0, \n        \"min\": 0.0, \n        \"max\": 0.0, \n        \"num_unique_values\": 1, \n        \"samples\": [\n          0.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"disgust\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 0.0, \n        \"min\": 0.0, \n        \"max\": 0.0, \n        \"num_unique_values\": 1, \n        \"samples\": [\n          0.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"predicted_emotion\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 1, \n        \"samples\": [\n          \"joy\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }\n  ],\n  \"type\": \"dataframe\"}

```

Step6:Analyze writing style through pos tagging and linguistic

```

from nltk import pos_tag
from nltk.tokenize import word_tokenize

```

Download the required NLTK resource

```

nltk.download("punkt")
nltk.download("average_perceptron_tagger")

```

```

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Error loading average_perceptron_tagger: Package
[nltk_data]   'average_perceptron_tagger' not found in index

```

False

POS Tagging Function

```

def get_pos_tags(text):
    tokens = word_tokenize(text)
    tags = pos_tag(tokens)
    return tags

```

```

print(df.columns)

Index(['review', 'sentiment', 'clean_review', 'label',
      'emotion_scores', 'joy',
      'anger', 'sadness', 'fear', 'surprise', 'disgust',
      'predicted_emotion'],
      dtype='object')

print(df.columns.tolist())

['review', 'sentiment', 'clean_review', 'label', 'emotion_scores',
'joy', 'anger', 'sadness', 'fear', 'surprise', 'disgust',
'predicted_emotion']

import nltk
from nltk import word_tokenize, pos_tag

nltk.download("punkt")
nltk.download("averaged_perceptron_tagger")

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!

True

def get_pos_tags(text):
    tokens = word_tokenize(str(text))
    return pos_tag(tokens)

import nltk

nltk.download("punkt")
nltk.download("punkt_tab")
nltk.download("averaged_perceptron_tagger_eng")

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.

True

import nltk

resources = [

```

```

    "punkt",
    "punkt_tab",
    "averaged_perceptron_tagger",
    "averaged_perceptron_tagger_eng",
    "wordnet",
    "omw-1.4",
    "stopwords"
]

for resource in resources:
    nltk.download(resource)

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!

nltk.download("averaged_perceptron_tagger_eng")

[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!

True

df["pos_tags"] = df["clean_review"].apply(get_pos_tags)

def pos_counts(tagged_words):
    counts = {
        "Nouns":0,
        "Verbs":0,
        "Adjectives":0,
        "Adverbs":0
    }

```

```

for word, tag in tagged_words:
    if tag.startswith("NN"):
        counts["Nouns"] += 1
    elif tag.startswith("VB"):
        counts["Verbs"] += 1
    elif tag.startswith("JJ"):
        counts["Adjectives"] += 1
    elif tag.startswith("RB"):
        counts["Adverbs"] += 1

return counts

df["pos_counts"] = df["pos_tags"].apply(pos_counts)

```

Convert Counts into Separate Columns

```

pos_df = df["pos_counts"].apply(pd.Series)

df = pd.concat([df, pos_df], axis=1)

```

Calculate Linguistic Ratios

Total words:

```

df["total_words"] = df["clean_review"].apply(lambda x: len(x.split()))

```

Noun Ratio

```

df["noun_ratio"] = df["Nouns"] / df["total_words"]

```

Verb Ratio

```

df["verb_ratio"] = df["Verbs"] / df["total_words"]

```

Adjective Ratio

```

df["adjective_ratio"] = df["Adjectives"] / df["total_words"]

```

Adverb Ratio

```

df["adverb_ratio"] = df["Adverbs"] / df["total_words"]

```

Display Results

```

df[[
    "clean_review",
    "Nouns",
    "Verbs",

```

```

    "Adjectives",
    "Adverbs",
    "noun_ratio",
    "verb_ratio",
    "adjective_ratio",
    "adverb_ratio"
  ]].head()

{"summary":{"\n  \"name\": \"\"]\\",\n  \"rows\": 5,\n  \"fields\": [\n  {\n    \"column\": \"clean_review\",\n    \"properties\": {\n  \"dtype\": \"string\",\n    \"num_unique_values\": 5,\n  \"samples\": [\n  \"awonderfullittleproductionbrbrthefilmingtechniqueisveryunassumingver  
yoldtimebbcfashionandgivesacomfortingandsometimesdiscomfortingsenseofr  
ealismtotheentirepiecebrbrtheactorsareextremelywellchosenmichaelsheenn  
otonlyhasgotallthepolaributhehasallthevoicesdownpattoooucantrulyseeth  
eseamlesseditingguidedbythereferencestowilliamsdiaryentriesnotonlyisit  
wellworththewatchingbutitisaterrificlywrittenandperformedpieceamasterf  
ulproductionaboutoneofthegreatmastersofcomedyandhislifebrbrtherealismr  
eallycomeshomewiththelittlethingsthefantasyoftheguardwhichratherthanu  
sethetraditionaldreamtechniquesremainssolidthendisappearsitplaysonourkn  
owledgeandoursensesparticularlywiththescenesconcerningortonandhalliwe  
landthesetsparticularlyoftheirflatwithhalliwellsmuralsdecoratingevery  
surfaceareterriblywelldone\\",\n  \"pettermatteisloveinthetimeofmoneyisavisuallystunningfilmtowatchmrm  
matteioffersusavividportraitabouthumanrelationsthisismoviethatseemstob  
ellinguswhatmoneypowerandsuccessdotopeopleinthedifferentsituationswe  
counterbrbrthisbeingavariationontheearthurschnitzlersplayaboutthesame  
themethedirectortransferstheactiontothepresenttimenewyorkwhereallthes  
edifferentcharactersmeetandconnecteachoneisconnectedinonewayoranothert  
tothenextpersonbutnooneseemstoknowthepreviouspointofcontactstylishly  
thefilmhasasophisticatedluxuriouslookwearetakentoseehowthesepeoplelive  
andtheworldtheyliveintheirownhabitatbrbrtheonlythingonegetsoutofallth  
sesoul sinthepictureisthedifferentstagesoflonelinesseachoneinhabitsabi  
gcityis notexactlythebestplaceinwhichhumanrelationsfindsincerefulfillm  
ent asone discernsisthecasewithmostofthepeopleweencounterbrbrtheactin  
g isgoodundermrmatteisdirectionstevebuscemirosariodawsoncarolkanemich  
aelimperiolia driangrenierandtherestofthetalentedcastmakethesecharac  
terscomealivebrbrwewishmrmatteigoodluckandawaitanxiouslyforhisnextwork\\",\n  \"ithoughtthiswasawonderfulwaytospendtimeonatoohotsummerweekends  
sitting intheairconditionedtheaterandwatchingalightheartedcomedythepl  
otissimpl isticbutthedialogueiswittyandthecharactersarelikableeventh  
ewellbreadspectedserialkillerwhilesomesmaybedisappointedwhentheyreal  
izethisisnotmatchpointriskaddictionithoughtitwasproofthatwoodyallen  
isstillfullyincontrolofthestylemanyofushavegrowntolovebrbrthiswasth  
emosttidlaughedatone ofwoodyscomediesinyearsdareisayadecadewhilei  
veneverbeenimpressedwithscarletjohansoninthisshemanagedtotonedown  
hersexyimageandjumpedrightintoaveragebutspiritedyoungwomanbrbrthi  
smaynotbethecrownjewelofhiscareerbutitwaswittierthandevilwearsprada  
andmoreinterestingthansupermanagreatcomedytogoseewithfriends\\",\n  ],\n  \"semantic_type\":
```

```

{"description": "Nouns", "properties": {"dtype": "number", "std": 0, "min": 1, "max": 1, "num_unique_values": 1, "samples": [1], "semantic_type": "noun"}, {"description": "Verbs", "properties": {"dtype": "number", "std": 0, "min": 0, "max": 0, "num_unique_values": 1, "samples": [0], "semantic_type": "verb"}, {"description": "Adjectives", "properties": {"dtype": "number", "std": 0, "min": 0, "max": 0, "num_unique_values": 1, "samples": [0], "semantic_type": "adjective"}, {"description": "Adverbs", "properties": {"dtype": "number", "std": 0, "min": 0, "max": 0, "num_unique_values": 1, "samples": [0], "semantic_type": "adverb"}, {"description": "noun_ratio", "properties": {"dtype": "number", "std": 0.0, "min": 1.0, "max": 1.0, "num_unique_values": 1, "samples": [1.0], "semantic_type": "ratio"}, {"description": "verb_ratio", "properties": {"dtype": "number", "std": 0.0, "min": 0.0, "max": 0.0, "num_unique_values": 1, "samples": [0.0], "semantic_type": "ratio"}, {"description": "adjective_ratio", "properties": {"dtype": "number", "std": 0.0, "min": 0.0, "max": 0.0, "num_unique_values": 1, "samples": [0.0], "semantic_type": "ratio"}, {"description": "adverb_ratio", "properties": {"dtype": "number", "std": 0.0, "min": 0.0, "max": 0.0, "num_unique_values": 1, "samples": [0.0], "semantic_type": "ratio"}], "type": "dataframe"}

```

Average Linguistic Ratios

```

print("Average Noun Ratio:", df["noun_ratio"].mean())
print("Average Verb Ratio:", df["verb_ratio"].mean())
print("Average Adjective Ratio:", df["adjective_ratio"].mean())
print("Average Adverb Ratio:", df["adverb_ratio"].mean())

```

Average Noun Ratio: 0.8789076681053608
Average Verb Ratio: 0.06750433625105885
Average Adjective Ratio: 0.030777298213061192
Average Adverb Ratio: 0.02006776652817555

Step7:Extract aspect specific sentiments using keywords and pattern-based method

Define Aspects and Keywords

```
aspects = {  
    "acting": ["acting", "actor", "actress", "cast", "performance"],  
    "story": ["story", "plot", "script", "screenplay"],  
    "music": ["music", "song", "soundtrack", "background score"],  
    "direction": ["director", "direction", "directing"],  
    "visuals": ["visual", "cinematography", "effects", "graphics",  
    "animation"]  
}
```

Define Sentiment Keywords

```
positive_words = [  
    "good", "great", "excellent", "amazing", "awesome",  
    "fantastic", "wonderful", "brilliant", "love", "liked",  
    "best", "perfect", "beautiful", "enjoyed"  
]  
  
negative_words = [  
    "bad", "poor", "boring", "terrible", "awful",  
    "worst", "hate", "disappointing", "weak",  
    "slow", "waste", "horrible"  
]
```

Create Aspect Sentiment Function

```
def aspect_sentiment(text):  
    text = text.lower()  
    result = {}  
    for aspect, keywords in aspects.items():  
        sentiment = "Not Mentioned"  
        for keyword in keywords:  
            if keyword in text:  
                # Find the position of the keyword
```

```

        index = text.find(keyword)

        # Look at nearby words (50 characters before and
after)
        context = text[max(0, index-50): index+50]

        if any(word in context for word in positive_words):
            sentiment = "Positive"

        elif any(word in context for word in negative_words):
            sentiment = "Negative"

        else:
            sentiment = "Neutral"

        break

    result[aspect] = sentiment

return result

```

Apply to the Dataset

```
df["aspect_sentiment"] = df["clean_review"].apply(aspect_sentiment)
```

Convert Dictionary into Columns

```

aspect_df = df["aspect_sentiment"].apply(pd.Series)
df = pd.concat([df, aspect_df], axis=1)

```

View Results

```

df[[
    "clean_review",
    "acting",
    "story",
    "music",
    "direction",
    "visuals"
]].head()

{"summary":{"\n  \"name\": \"\"]\n",\n  \"rows\": 5,\n  \"fields\": [\n  {\n    \"column\": \"clean_review\",
    \"properties\": {\n
    \"dtype\": \"string\",
    \"num_unique_values\": 5,\n
    \"samples\": [\n
    \"awonderfullittleproductionbrbrthefilmingtechniqueisveryunassumingver
yoldtimebbcfashionandgivesacomfortingandsometimesdiscomfortingsenseofr
ealismtotheentirepiecebrbrtheactorsareextremelywellchosenmichaelsheenn
otonlyhasgotallthepolaributhehasallthevoicesdownpattooyoucantrulyseeth
eseamlesseditingguidedbythereferencestowilliamsdiaryentriesnotonlyisit

```

wellworththewatchingbutitisaterrificlywrittenandperformedpieceamasterful production about one of the great masters of comedy and his life brbr the realism really comes home with the little things the fantasy of the guard which rather than use the traditional dream techniques remain solid then disappears it plays on our knowledge and our senses particularly with the scenes concerningorton and halliwell and the sets particularly of their flat with halliwell's murals decorating every surface are terribly well done",\n

\n"pettermatteisloveinthetimeofmoneyisavisuallystunningfilmtowatchmrmatteioffersusavividportraitabout human relations this is a movie that seems to be telling us what money power and success do to people in the different situations between counterbrbr this being a variation on the earth schnitzlers play about the same theme the director transfers the action to the present time new york where all these different characters meet and connect each one is connected in one way or another to the next person but no one seems to know the previous point of contact stylishly the film has a sophisticated luxurious look we are taken to see how these people live and the world they live in their own habitat brbr the only thing one gets out of all these soul sin the picture is the different stages of loneliness each one inhabits a big city is not exactly the best place in which human relations find sincere fulfillment as one discerns is the case with most of the people we encounter brbr the acting is good under mrmatteis direction steve buscemi rosario dawson carol kanemichael imperioli ariana grenier and the rest of the talented cast make these characters come alive brbr we wish mrmattei good luck and await anxiously for his next work",\n

\n"i thought this was a wonderful way to spend time on a too hot summer weekend sitting in the air conditioned theater and watching a light hearted comedy the plot is simplistic but the dialogue is witty and the characters are likable even though well bread suspected serial killer while some may be disappointed when they realize this is not much of a point risk addiction i thought it was proof that woody allen is still fully in control of the style many of us have grown to love brbr this was the most idlaughed at one of woody's comedies in years dare i say a decade while i've never been impressed with scarlet johanson in this she managed to tone down her sexy image and jumped right into an average but spirited young woman brbr this may not be the crown jewel of his career but it was wittier than devil wears prada and more interesting than superman a great comedy to go see with friends",\n

```
],\n  \"semantic_type\":\n  \"\", \n  \"description\": \"\", \n  },\n  {\n  \"column\": \"acting\", \n  \"properties\": {\n  \"dtype\":\n  \"string\", \n  \"num_unique_values\": 3, \n  \"samples\":\n  [\n  \"Not Mentioned\", \n  \"Neutral\", \n  \"Positive\", \n  ], \n  \"semantic_type\": \"\", \n  \"description\": \"\", \n  }, \n  {\n  \"column\":\n  \"story\", \n  \"properties\": {\n  \"dtype\": \"category\", \n  \"num_unique_values\": 2, \n  \"samples\": [\n  \"Neutral\", \n  \"Not Mentioned\", \n  ], \n  \"semantic_type\": \"\", \n  \"description\": \"\", \n  }, \n  {\n  \"column\": \"music\", \n  \"properties\": {\n  \"dtype\": \"category\", \n  \"num_unique_values\": 1, \n  \"samples\": [\n  \"Not Mentioned\", \n  ], \n  \"semantic_type\": \"\", \n  \"description\": \"\", \n  }, \n  {\n  \"column\": \"direction\", \n  \"properties\": {\n  \"dtype\": \"category\", \n
```

```

{"num_unique_values": 2, "samples": [{"Neutral", "semantic_type": "", "description": "", "column": "visuals", "properties": {"dtype": "category", "num_unique_values": 2, "samples": [{"Positive", "semantic_type": "", "description": ""}]}], "type": "dataframe"}

```

Srep8:Generate contextual embedding using a pretrain BERT model

Install Transformers

```
pip install transformers torch
```

```

Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.13.1)
Requirement already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.23.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.26.8)
Requirement already satisfied: safetensors>=0.8.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch) (3.29.7)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.16.0)
Requirement already satisfied: setuptools<82 in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)

```

Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (8.4.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (1.5.1)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (0.28.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(1.5.4)
Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(0.0.4)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (4.14.2)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (2026.6.17)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0-
>transformers) (3.18)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer-
>transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer-
>transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in

```
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-  
>rich>=13.8.0->typer->transformers) (0.1.2)
```

Import Libraries

```
import torch  
import pandas as pd  
  
from transformers import BertTokenizer, BertModel
```

Load the Pretrained BERT Model

```
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")  
  
bert_model = BertModel.from_pretrained("bert-base-uncased")
```

Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.

```
{"model_id":"2387bc51be3f4caa9d50280c8b7fe112","version_major":2,"vers  
ion_minor":0}
```

```
{"model_id":"8830094a9d8f41338486a34e0d0e9c47","version_major":2,"vers  
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```

```
{"model_id":"782dd6c313064f0788a6fd00aed2177a","version_major":2,"vers  
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```

```
{"model_id":"95806bc78fea4cb095aa4bedb5f7dfa1","version_major":2,"vers  
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{"model_id":"b48f651e4e85411a984a0429c652ba22","version_major":2,"vers  
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```

```
{"model_id":"c22bbb2f43f94945b9a15c9a6466e8c8","version_major":2,"vers  
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```

```
{"model_id":"962840f712404d049d84ff2bd0a6bd14","version_major":2,"vers  
ion_minor":0}
```

[transformers] BertModel LOAD REPORT from: bert-base-uncased

Key	Status		
cls.predictions.transform.dense.bias	UNEXPECTED		
cls.predictions.bias	UNEXPECTED		
cls.predictions.transform.LayerNorm.bias	UNEXPECTED		
cls.predictions.transform.LayerNorm.weight	UNEXPECTED		
cls.seq_relationship.bias	UNEXPECTED		

cls.seq_relationship.weight	UNEXPECTED	
cls.predictions.transform.dense.weight	UNEXPECTED	

Notes:

- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Create a Function to Generate BERT Embeddings

```
def get_bert_embedding(text):
    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding="max_length",
        max_length=128
    )

    with torch.no_grad():
        outputs = bert_model(**inputs)

    # CLS token embedding
    embedding = outputs.last_hidden_state[:, 0, :]

    return embedding.squeeze().numpy()
```

Generate Embeddings

```
df["bert_embedding"] =
df["clean_review"].head(5).apply(get_bert_embedding)

embeddings = []

for review in df["clean_review"].head(5):
    embeddings.append(get_bert_embedding(review))
```

Check the Embedding Shape

```
print(embeddings[0].shape)

(768,)
```

View the Embeddings

```
print(embeddings[0])

[-5.35640180e-01  5.02438009e-01 -5.77307224e-01 -3.27018291e-01
 -4.72045839e-01 -1.38240367e-01  8.16586316e-01  5.16603053e-01
 -5.96357808e-02 -3.08990367e-02 -4.37952220e-01 -4.60927099e-01
 1.03270933e-01 -4.62508164e-02  4.28058088e-01 -2.67332971e-01]
```

1.23826023e-02	3.70137095e-01	7.87010550e-01	-2.34458800e-02
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-2.04642490e-01	-1.04893133e-01	-5.55921197e-02	-3.39133710e-01
9.63620916e-02	-6.79817870e-02	8.19018260e-02	5.08137941e-01
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2.46688002e-03	-1.40747517e-01	-1.47699997e-01	3.32861096e-01
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-1.52390152e-01	-7.09142089e-01	-2.16115594e+00	-4.15104181e-01
-5.55044413e-01	-2.99349397e-01	3.47789712e-02	1.72865301e-01
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9.92188454e-02	-2.63161516e+00	3.49052399e-01	-1.01522328e-02
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```

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Save Embeddings in the DataFrame

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eseamlesseditingguidedbythereferencestowilliamsdiaryentriesnotonlyisit
wellworththewatchingbutitisaterrificlywrittenandperformedpieceamasterf
ulproductionaboutoneofthegreatmastersofcomedyandhislifebrbrtherealismr
eallycomeshomewiththelittlethingsthefantasyoftheguardwhichratherthanu
sethetraditionaldreamtechniquesremainssolidthendisappearsitplaysonourkn
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llandthesetsparticularlyoftheirflatwithhalliwellsmuralsdecoratingevery
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teioffersusavividportraitabouthumanrelationsthisisamoviehatseemstobet
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counterbrbrthisbeingavariationontheearthurschnitzlersplayaboutthesameth
emethedirectortransferstheactiontothepresenttimenewyorkwhereallthesedi
fferentcharactersmeetandconnecteachoneisconnectedinonewayoranothertoth
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mhasasophisticatedluxuriouslookwearetakentoseehowthesepeopleliveandthe
worldtheyliveintheirownhabitatbrbrtheonlythingonegetsoutofallthesesoul
sinthepictureisthedifferentstagesoflonelinesseachoneinhabitsabigcityis
notexactlythebestplaceinwhichhumanrelationsfindsincerefulfillmentasone
discernsisthecasewithmostofthepeopleweencounterbrbrtheactingisgoodunde
rmrmmatteisdirectionstevebuscemirosariodawsoncarolkanemichaelimperiolia
driangrenierandtherestofthetalentedcastmakethesecharacterscomealivebrb
rwewishmrmatteigoodluckandawaitanxiouslyforhisnextwork\", \n
\"ithoughtthiswasawonderfulwaytospendtimeonatoohotsummerweekendsitting
intheairconditionedtheaterandwatchingalightheartedcomedytheplotissimpl
isticbutthedialogueiswittyandthecharactersarelikableeventhewellbreadsu
spectedserialkillerwhilesomemaybedisappointedwhentheyrealizethisisnotm
atchpointriskaddictionithoughtitwasproofthatwoodyallenisstillfullyinco
ntrolofthestylemanyofushavegrowntolovebrbrthiswasthemostidlaughedatone
ofwoodyscomediesinyearsdareisayadecadewhileiiveneverbeenimpressedwiths
carletjohansoninthisshemanagedtotonedownhersexyimageandjumpedrightintoa
averagebutspiritedyoungwomanbrbrthismaynotbethecrownjewelofhiscareerbu
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